University of Toronto
Combustion Processes (AER315) **Handout #2**
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Heat and Mass Transfer

Outline *In this handout, we will look at the another important component of combustion processes, i.e. transport phenomenon. After we determine the product concentration and the energy released from these processes, we look at how mass and energy is transported in temporal and spatial scales. This is done using the concepts of heat and mass transfer. We first provide a rudimentary understanding of transport processes and a molecular description of mass transfer. We then notice the similarities in heat and mass transfer processes and finally study the application of these concepts to important practical systems.*

Contents

1 Mass Transfer

1.1 Fick's Law of Diffusion

In a binary system with 2 chemical species, species A always diffuses in the direction from higher concentration of A to low concentration of A. The same is true for species B. The binary mass diffusivity is a property of the mixture and can be expressed by either $\mathcal{D}_{\mathcal{AB}}$ or $\mathcal{D}_{\mathcal{BA}}$ respectively for A diffusing into B and B diffusing into A. Units of $\mathcal{D}_{\mathcal{AB}}$ or $\mathcal{D}_{\mathcal{BA}}$ is m^2/s. Then, Fick's law of diffusion, written in terms of mass diffusion flux $\dot{m}''_{\mathrm{A,diff}}$ for the binary system, is

$$\dot{m}''_{\mathrm{A,diff}} = -\rho\mathcal{D}_{\mathrm{AB}}\frac{\mathrm{d}Y_{\mathrm{A}}}{\mathrm{d}x} \tag{1}$$

where $\dot{m}''_{\mathrm{A,diff}}$ is defined as the mass flowrate through diffusion of species A per unit area perpendicular to the flow , i.e $\dot{m}''_{\mathrm{A,diff}} = \dot{m}_{\mathrm{A,diff}}/A$ where A is the area perpendicular to the flow. $\dot{m}''_{\mathrm{A,diff}}$ has the units kg/(s m^2). Note that the negative sign causes the flux to be positive in the x-direction when the concentration gradient is negative. Now, relative to the stationary coordinate, the overall flux associated with species A is given by

$$\underbrace{\dot{m}''_{\mathrm{A}}}_{\substack{\text{mass flow of A}\\ \text{per unit area}}} = \underbrace{Y_{\mathrm{A}}(\dot{m}''_{\mathrm{A}} + \dot{m}''_{\mathrm{B}})}_{\substack{\text{mass flow of A}\\ \text{associated with bulk}\\ \text{flow per unit area}}} - \underbrace{\rho\mathcal{D}_{\mathrm{AB}}\frac{\mathrm{d}Y_{\mathrm{A}}}{\mathrm{d}x}}_{\substack{\text{mass flow of A}\\ \text{associated with}\\ \text{molecular diffusion}}} \tag{2}$$

Here, species A is transported by two means. Firstly, through the bulk motion of the fluid, and secondly, through molecular diffusion. Therefore, the diffusion mass flux $\dot{m}''_{\mathrm{A,diff}}$ from (1) has been replaced with the overall mass flux of species A given by $\dot{m}''_{\mathrm{A}}$ in (2). A more general form of (2) is given by

$$\overrightarrow{\dot{m}''_{\mathrm{A}}} = Y_{\mathrm{A}}(\overrightarrow{\dot{m}''_{\mathrm{A}}} + \overrightarrow{\dot{m}''_{\mathrm{B}}}) - \rho\mathcal{D}_{\mathrm{AB}}\nabla Y_{\mathrm{A}} \tag{3}$$

where symbols with over arrows represent vector quantities. (3) can also be written in a molar form given by:

$$\overrightarrow{\dot{N}''_{\mathrm{A}}} = \chi_{\mathrm{A}}(\overrightarrow{\dot{N}''_{\mathrm{A}}} + \overrightarrow{\dot{N}''_{\mathrm{B}}}) - c\mathcal{D}_{\mathrm{AB}}\nabla \chi_{\mathrm{A}} \tag{4}$$

where $\dot{N}''_{\mathrm{A}}$, (kmol/(s m^2), is the molar flux of species A; χ_{A} is mole fraction, and c is the molar concentration, kmol/m^3. Meanings of bulk flow and diffusional flux can be better explained if we consider the complete mixture so that:

$$\underbrace{\dot{m}''}_{\substack{\text{mixture}\\ \text{mass flux}}} = \underbrace{\dot{m}''_{\mathrm{A}}}_{\substack{\text{species A}\\ \text{mass flux}}} + \underbrace{\dot{m}''_{\mathrm{B}}}_{\substack{\text{species B}\\ \text{mass flux}}} \tag{5}$$

If we substitute for individual species fluxes from (2) into (5), we get:

$$\begin{aligned}\dot{m}'' &= Y_{\mathrm{A}}\dot{m}'' - \rho\mathcal{D}_{\mathrm{AB}}\frac{\mathrm{d}Y_{\mathrm{A}}}{\mathrm{d}x} + Y_{\mathrm{B}}\dot{m}'' - \rho\mathcal{D}_{\mathrm{BA}}\frac{\mathrm{d}Y_{\mathrm{B}}}{\mathrm{d}x}\\ &= (Y_{\mathrm{A}} + Y_{\mathrm{B}})\dot{m}'' - \rho\mathcal{D}_{\mathrm{AB}}\frac{\mathrm{d}Y_{\mathrm{A}}}{\mathrm{d}x} - \rho\mathcal{D}_{\mathrm{BA}}\frac{\mathrm{d}Y_{\mathrm{B}}}{\mathrm{d}x}\end{aligned} \tag{6}$$

For a binary mixture, $Y_{\mathrm{A}}+Y_{\mathrm{B}}=1$; then:

$$-\underbrace{\rho\mathcal{D}_{\mathrm{AB}}\frac{\mathrm{d}Y_{\mathrm{A}}}{\mathrm{d}x}}_{\substack{\text{diffusional flux}\\ \text{of species A}}}-\underbrace{\rho\mathcal{D}_{\mathrm{BA}}\frac{\mathrm{d}Y_{\mathrm{B}}}{\mathrm{d}x}}_{\substack{\text{diffusional flux}\\ \text{of species B}}}=0 \tag{7}$$

Some Some cautionary remarks for the above analysis are given below

- We are assuming a binary gas and the diffusion is a result of concentration gradients only (ordinary diffusion).
- Gradients of temperature and pressure can produce species diffusion.
- Soret effect: species diffusion as a result of temperature gradient.
- In most combustion systems, these effects are small and can be neglected.

1.2 Molecular Basis for Diffusion

Here, we apply some concepts from the kinetic theory of gases to understand the process of molecular diffusion. Consider a stationary (no bulk flow) plane layer of a binary gas mixture consisting of rigid, non-attracting molecules. Molecular masses of A and B are identical. A concentration (mass-fraction) gradient exists in x-direction, and is sufficiently small that over smaller distances the gradient can be assumed to be linear. Average molecular properties from kinetic theory of gases:

$$\bar{v}\equiv \substack{\text{Mean speed}\\ \text{of species A}\\ \text{molecules}}=\left(\frac{8k_BT}{\pi m_{\mathrm{A}}}\right)^{1/2} \tag{8a}$$

$$Z''_{\mathrm{A}}\equiv \substack{\text{Wall collision}\\ \text{frequency of A}\\ \text{molecules per unit area}}=\frac{1}{4}\left(\frac{n_{\mathrm{A}}}{V}\right)\bar{v} \tag{8b}$$

$$\lambda\equiv \text{Mean free path}=\frac{1}{\sqrt{2}\pi(n_{\mathrm{tot}}/V)\sigma^2} \tag{8c}$$

$$a\equiv \substack{\text{Average perpendicular distance}\\ \text{from plane of last collision to}\\ \text{plane where next collision occurs}}=\frac{2}{3}\lambda \tag{8d}$$

where k_B is Boltzmann's constant, m_{A} is the mass of a single A molecule, (n_{A}/V) is number of A molecules per unit volume, (n_{tot}/V) is total number of molecules per unit volume and σ is the diameter of both A and B molecules. Net flux of A molecules at the x-plane:

$$\dot{m}''_{\mathrm{A}}=\dot{m}''_{\mathrm{A},(+)x-\mathrm{dir}}-\dot{m}''_{\mathrm{A},(-)x-\mathrm{dir}} \tag{9}$$

In terms of collision frequency, (9) becomes

$$\underbrace{\dot{m}''_{\mathrm{A}}}_{\substack{\text{Net mass}\\ \text{flux of}\\ \text{species A}}}=\underbrace{(Z''_{\mathrm{A}})_{x-a}}_{\substack{\text{Number of A}\\ \text{crossing plane x}\\ \text{originating from}\\ \text{plane at (x−a)}}}m_{\mathrm{A}}-\underbrace{(Z''_{\mathrm{A}})_{x+a}}_{\substack{\text{Number of A}\\ \text{crossing plane x}\\ \text{originating from}\\ \text{plane at (x+a)}}}m_{\mathrm{A}} \tag{10}$$

Since $\rho\equiv m_{\mathrm{tot}}/V_{\mathrm{tot}}$, then we can relate Z''_{A} to mass fraction, Y_{A} (from (8b)).

$$Z''_{\mathrm{A}}m_{\mathrm{A}}=\frac{1}{4}\frac{n_{\mathrm{A}}m_{\mathrm{A}}}{m_{\mathrm{tot}}}\rho\bar{v}=\frac{1}{4}Y_{\mathrm{A}}\rho\bar{v} \tag{11}$$

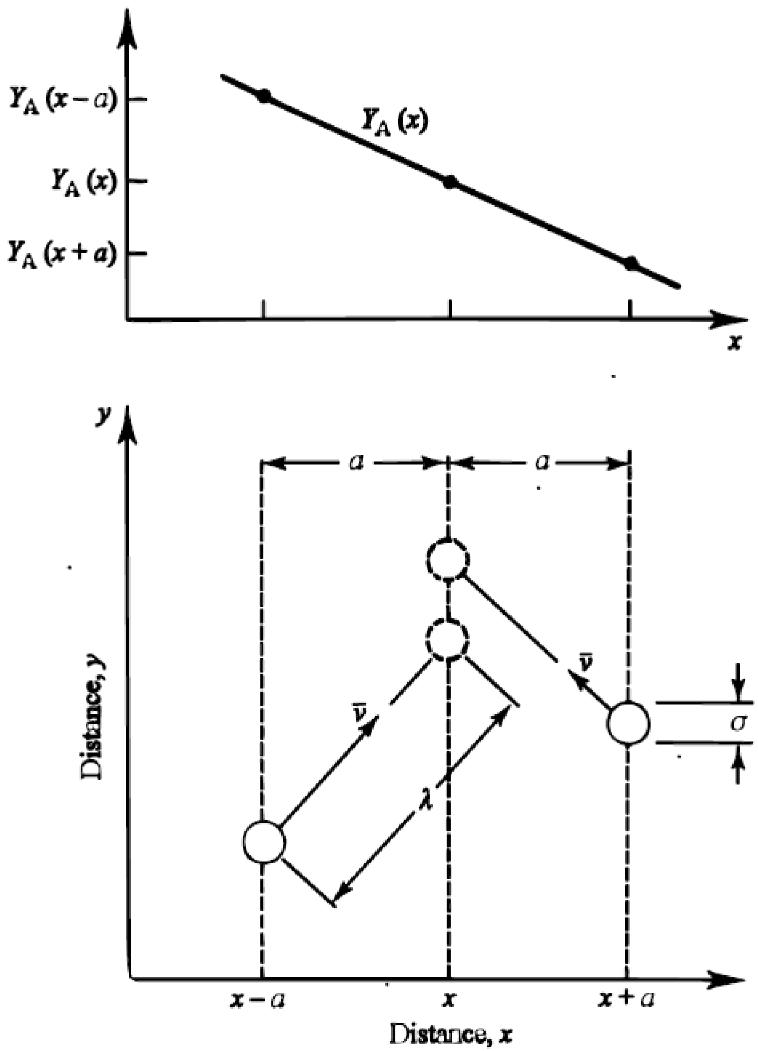

Figure 1: **Top**: Linear mass fraction gradient over small distance x-a to x+a. **Bottom**: Description of average molecular properties from kinetic theory of gases.

Substituting (11) into (10)

$$\dot{m}''_{\mathrm{A}} = \frac{1}{4}\rho\bar{v}(Y_{\mathrm{A},x-a} - Y_{\mathrm{A},x+a}) \tag{12}$$

With linear concentration assumption:

$$\frac{\mathrm{d}Y}{\mathrm{d}x} = \frac{Y_{\mathrm{A},x-a} - Y_{\mathrm{A},x+a}}{2a} = \frac{Y_{\mathrm{A},x-a} - Y_{\mathrm{A},x+a}}{4\lambda/3} \tag{13}$$

From the last two equations, we get

$$\dot{m}''_{\mathrm{A}} = -\rho\frac{\bar{v}\lambda}{3}\frac{\mathrm{d}Y_{\mathrm{A}}}{\mathrm{d}x} \tag{14}$$

Comparing (14) with (1), $\mathcal{D}_{\mathrm{AB}}$ is

$$\mathcal{D}_{\mathrm{AB}} = \bar{v}\lambda/3 \tag{15}$$

Substituting for $\bar{v}$ and λ, along with ideal-gas equation of state, $PV = nk_BT$

$$\mathcal{D}_{\mathrm{AB}} = \frac{2}{3}\left(\frac{k_B^3 T}{\pi^3 m_{\mathrm{A}}}\right)^{1/2}\frac{T}{\sigma^2 P} \tag{16a}$$

$$\mathcal{D}_{\mathrm{AB}} \propto T^{3/2}P^{-1} \tag{16b}$$

Diffusivity strongly depends on temperature and is inversely proportional to pressure. Mass flux of species A, however, depends on $\rho\mathcal{D}_{\mathrm{AB}}$, which then gives:

$$\rho\mathcal{D}_{\mathrm{AB}} \propto T^{1/2}P^0 = T^{1/2} \tag{17}$$

In some practical/simple combustion calculations, the weak temperature dependence is neglected and $\rho\mathcal{D}$ is treated as a constant.

1.3 Comparison to Heat Conduction

Fourier's law of heat conduction:

$$\dot{Q}''_x = -k\frac{\mathrm{d}T}{\mathrm{d}x} \tag{18}$$

Both expressions indicate a flux ($\dot{m}''_{\mathrm{A,diff}}$ or $\dot{Q}''_x$) being proportional to the gradient of a scalar quantity [$(\mathrm{d}Y_{\mathrm{A}}/\mathrm{d}x)$ or $(\mathrm{d}T/\mathrm{d}x)$]. We apply the same kinetic theory concepts to the transport of energy. Same assumptions as in the molecular diffusion case are taken with $\bar{v}$ and λ having the same definitions. Molecular collision frequency is now based on the total number density of molecules, n_{tot}/V,

$$Z'' = \substack{\text{Average wall collision}\\ \text{frequency per unit area}} = \frac{1}{4}\left(\frac{n_{\mathrm{tot}}}{V}\right)\bar{v} \tag{19}$$

In the no-interaction-at-a-distance hard-sphere model of the gas, the only energy storage mode is molecular translational (kinetic) energy. Energy balance at the x-plane is then given by:

$$\substack{\text{Net energy flow}\\ \text{in } x\text{−direction}} = \substack{\text{kinetic energy flux}\\ \text{with molecules}\\ \text{from } x-a \text{ to } x} - \substack{\text{kinetic energy flux}\\ \text{with molecules}\\ \text{from } x+a \text{ to } x}$$

$$\dot{Q}''_x = Z''(ke)_{x-a} - Z''(ke)_{x+a} \tag{20}$$

ke is given by

$$ke = \frac{1}{2}m\bar{v}^2 = \frac{3}{2}k_BT \tag{21}$$

heat flux can be related to temperature as

$$\dot{Q}''_x = \frac{3}{2}k_B Z''(T_{x-a} - T_{x+a}) \tag{22}$$

The temperature gradient

$$\frac{\mathrm{d}T}{\mathrm{d}x} = \frac{T_{x+a} - T_{x-a}}{2a} \tag{23}$$

Substituting (23) into (22), and definitions of Z'' and a

$$\dot{Q}''_x = -\frac{1}{2} k_B \Big(\frac{n}{V}\Big) \bar{v} \lambda \frac{\mathrm{d}T}{\mathrm{d}x} \tag{24}$$

Comparing to Eqn. 3.4, k is

$$k = \frac{1}{2} k_B \Big(\frac{n}{V}\Big) \bar{v} \lambda \tag{25}$$

In terms of T and molecular size,

$$k = \Big(\frac{k_B^3}{\pi^3 m \sigma^4}\Big)^{1/2} T^{1/2} \tag{26}$$

Dependence of k on T (similar to $\rho\mathcal{D}$)

$$k \propto T^{1/2} \tag{27}$$

It is worthwhile to note that for real gases, T dependency is larger.

1.4 Species Conservation

Here we will write the expression for species conservation in a 1D control volume. Species A flows into and out of the control volume as a result of the combined action of bulk flow and diffusion. Within the control volume, species A may be created or destroyed as a result of chemical reaction. The net rate of increase in the mass of A within the control volume relates to the mass fluxes and reaction rate as follows:

$$\underbrace{\frac{\mathrm{d}m_{\mathrm{A},cv}}{\mathrm{d}t}}_{\substack{\text{Rate of increase}\\ \text{of mass A}\\ \text{within CV}}} = \underbrace{[\dot{m}''_{\mathrm{A}} A]_x}_{\substack{\text{Mass flow}\\ \text{of A into}\\ \text{CV}}} - \underbrace{[\dot{m}''_{\mathrm{A}} A]_{x+\Delta x}}_{\substack{\text{Mass flow}\\ \text{of A out}\\ \text{of the CV}}} + \underbrace{\dot{m}'''_{\mathrm{A}} V}_{\substack{\text{Mass prod.}\\ \text{rate of A}\\ \text{by reaction}}} \tag{28}$$

where $\dot{m}'''_{\mathrm{A}}$ is the mass production rate of species A per unit volume and $\dot{m}''_{\mathrm{A}}$ is defined by (2). Within the control volume $m_{\mathrm{A},cv} = Y_{\mathrm{A}} m_{cv} = Y_{\mathrm{A}} \rho V_{cv}$, and the volume $V_{cv} = A \cdot \Delta x$;(28)

$$A\Delta x \frac{\partial(\rho Y_{\mathrm{A}})}{\partial t} = A\Big[Y_{\mathrm{A}} \dot{m}'' - \rho\mathcal{D}_{\mathrm{AB}} \frac{\partial Y_{\mathrm{A}}}{\partial x}\Big]_x - A\Big[Y_{\mathrm{A}} \dot{m}'' - \rho\mathcal{D}_{\mathrm{AB}} \frac{\partial Y_{\mathrm{A}}}{\partial x}\Big]_{x+\Delta x} + \dot{m}'''_{\mathrm{A}} A\Delta x \tag{29}$$

Dividing by $A\Delta x$ and taking limit as $\Delta x \to 0$,

$$\frac{\partial(\rho Y_{\mathrm{A}})}{\partial t} = -\frac{\partial}{\partial x}\Big[Y_{\mathrm{A}} \dot{m}'' - \rho\mathcal{D}_{\mathrm{AB}} \frac{\partial Y_{\mathrm{A}}}{\partial x}\Big] + \dot{m}'''_{\mathrm{A}} \tag{30}$$

For the steady-flow,

$$\dot{m}'''_{\mathrm{A}} - \frac{\mathrm{d}}{\mathrm{d}x}\Big[Y_{\mathrm{A}} \dot{m}'' - \rho\mathcal{D}_{\mathrm{AB}} \frac{\mathrm{d}Y_{\mathrm{A}}}{\mathrm{d}x}\Big] = 0 \tag{31}$$

(31) is the steady-flow, one-dimensional form of species conservation for a binary gas mixture. For a multidimensional case, (31) can be generalized as

$$\underbrace{\dot{m}'''_{\mathrm{A}}}_{\substack{\text{Net rate of}\\ \text{species A production}\\ \text{by chemical reaction}}} - \underbrace{\nabla \cdot \overrightarrow{\dot{m}''_{\mathrm{A}}}}_{\substack{\text{Net flow of species}\\ \text{A out of}\\ \text{control volume}}} = 0 \tag{32}$$

2 Applications of Mass Transfer

2.1 Stefan Problem

Consider liquid A, maintained at a fixed height in a glass cylinder as illustrated in Fig.2 . A mixture of gas A and gas B fl ow across the top of the cylinder. If the concentration of A in the fl owing gas is less than the concentration of A at the liquid–vapor interface, a driving force for mass transfer exists and species A will diffuse from the liquid–gas interface to the open end of the cylinder. The assumptions are listed below

- Liquid A in the cylinder maintained at a fixed height.
- Steady-state
- [A] in the flowing gas is less than [A] at the liquid-vapour interface.
- B is insoluble in liquid A

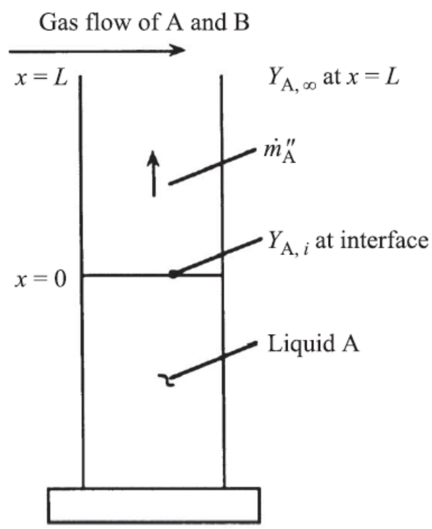

Figure 2: Diffusion of vapor A through a stagnant column of gas B, i.e., the Stefan problem.

Overall conservation of mass:

$$\dot{m}''(x) = \text{constant} = \dot{m}''_{\mathrm{A}} + \dot{m}''_{\mathrm{B}} \tag{33}$$

Since $\dot{m}''_{\mathrm{B}} = 0$, then

$$\dot{m}''_{\mathrm{A}} = \dot{m}''(x) = \text{constant} \tag{34}$$

Then, (2) now becomes

$$\dot{m}''_{\mathrm{A}} = Y_{\mathrm{A}}\dot{m}''_{\mathrm{A}} - \rho\mathcal{D}_{\mathrm{AB}}\frac{\mathrm{d}Y_{\mathrm{A}}}{\mathrm{d}x} \tag{35}$$

Rearranging and separating variables

$$-\frac{\dot{m}''_{\mathrm{A}}}{\rho\mathcal{D}_{\mathrm{AB}}}\mathrm{d}x = \frac{\mathrm{d}Y_{\mathrm{A}}}{1-Y_{\mathrm{A}}} \tag{36}$$

Assuming $\rho\mathcal{D}_{\mathrm{AB}}$ to be constant, integrate (36)

$$-\frac{\dot{m}''_{\mathrm{A}}}{\rho\mathcal{D}_{\mathrm{AB}}}x = -\ln[1-Y_{\mathrm{A}}] + C \tag{37}$$

With the boundary condition

$$Y_{\mathrm{A}}(x=0) = Y_{\mathrm{A},i} \tag{38}$$

We can eliminate C; then

$$Y_{\mathrm{A}}(x) = 1 - (1-Y_{\mathrm{A},i})\exp\Big[\frac{\dot{m}''_{\mathrm{A}}x}{\rho\mathcal{D}_{\mathrm{AB}}}\Big] \tag{39}$$

The mass flux of A, $\dot{m}''_{\mathrm{A}}$, can be found by letting $Y_{\mathrm{A}}(x=L) = Y_{\mathrm{A},\infty}$. Then, (39) reads

$$\dot{m}''_{\mathrm{A}} = \frac{\rho\mathcal{D}_{\mathrm{AB}}}{L}\ln\Big[\frac{1-Y_{\mathrm{A},\infty}}{1-Y_{\mathrm{A},i}}\Big] \tag{40}$$

Mass flux is proportional to $\rho\mathcal{D}$, and inversely proportional to L.

2.2 Liquid–Vapor Interface Boundary Conditions

In the example above, we treated the gas-phase mass fraction of the diffusing species at the liquid–vapor interface, $Y_{\mathrm{A,i}}$, as a known quantity. Unless this mass fraction is measured, which is unlikely, some means must be found to calculate or estimate its value. This can be done by assuming equilibrium exists between the liquid and vapor phases of species A. With this equilibrium assumption, and the assumption of ideal gases, the partial pressure of species A on the gas side of the interface must equal the saturation pressure associated with the temperature of the liquid, i.e.,

$$P_{\mathrm{A},i} = P_{\mathrm{sat}}(T_{\mathrm{liq},i}) \tag{41}$$

Partial pressure can be related to mole fraction and mass fraction

$$Y_{\mathrm{A},i} = \frac{P_{\mathrm{sat}}(T_{\mathrm{liq},i})}{P}\frac{MW_{\mathrm{A}}}{MW_{\mathrm{mix},i}} \tag{42}$$

To find $Y_{\mathrm{A},i}$ we need to know the interface temperature. In crossing the liquid-vapour boundary, we maintain continuity of temperature

$$T_{\mathrm{liq},i}(x=0^-) = T_{\mathrm{vap},i}(x=0^+) = T(0) \tag{43}$$

and energy is conserved at the interface. Heat is transferred from gas to liquid, $\dot{Q}_{g-i}$. Some of this heats the liquid, $\dot{Q}_{i-l}$, while the remainder causes phase change

$$\dot{Q}_{g-i} - \dot{Q}_{i-l} = \dot{m}(h_{\mathrm{vap}} - h_{\mathrm{liq}}) = \dot{m}h_{\mathrm{fg}} \tag{44}$$

or

$$\dot{Q}_{\mathrm{net}} = \dot{m}h_{\mathrm{fg}} \tag{45}$$

(45) can be used to calculate the net heat transferred to the interface if the evaporation rate, $\dot{m}$, is known. Conversely, if $\dot{Q}_{net}$ is known, the evaporation rate can be determined.

2.3 Droplet Evaporation

The problem of the evaporation of a single liquid droplet in a quiescent environment is just the Stefan problem for a spherically symmetric coordinate system. The assumptions are given by:

- The evaporation process is quasi-steady.
- The droplet temperature is uniform, and the temperature is assumed to be some fixed value below the boiling point of the liquid.
- The mass fraction of vapour at the droplet surface is determined by liquid-vapour equilibrium at the droplet temperature.
- We assume constant physical properties, e.g., $\rho\mathcal{D}$.

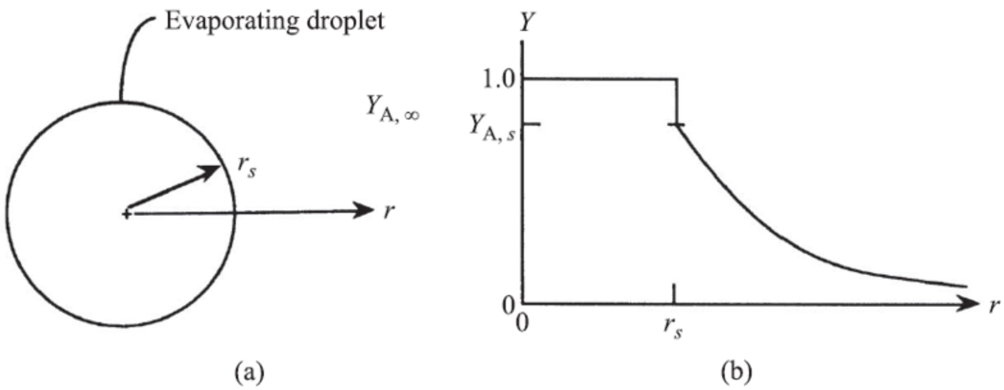

Figure 3: Evaporation of a liquid droplet in a quiescent environment

Overall mass conservation:

$$\dot{m}(r) = \text{constant} = 4\pi r^2 \dot{m}'' \tag{46}$$

Species conservation for the droplet vapour:

$$\dot{m}''_{\text{A}} = Y_{\text{A}}\dot{m}''_{\text{A}} - \rho\mathcal{D}_{\text{AB}}\frac{\mathrm{d}Y_{\text{A}}}{\mathrm{d}r} \tag{47}$$

Substitute (46) into (47) and solve for $\dot{m}$,

$$\dot{m} = -4\pi r^2 \frac{\rho\mathcal{D}_{\text{AB}}}{1 - Y_{\text{A}}}\frac{\mathrm{d}Y_{\text{A}}}{\mathrm{d}r} \tag{48}$$

Integrating and applying the boundary condition

$$Y_{\text{A}}(r = r_s) = Y_{\text{A},s} \tag{49}$$

yields

$$Y_{\text{A}}(r) = 1 - \frac{(1 - Y_{\text{A},s})\exp\left[-\dot{m}/(4\pi\rho\mathcal{D}_{\text{AB}}r)\right]}{\exp\left[-\dot{m}/(4\pi\rho\mathcal{D}_{\text{AB}}r_s)\right]} \tag{50}$$

Evaporation rate can be determined from (50) by letting $Y_\mathrm{A} = Y_{\mathrm{A},\infty}$ for $r \to \infty$:

$$\dot{m} = 4\pi r_s \rho \mathcal{D}_\mathrm{AB} \ln\left[\frac{(1 - Y_{\mathrm{A},\infty})}{(1 - Y_{\mathrm{A},s})}\right] \tag{51}$$

In (51), we can define the dimensionless transfer number, B_Y,

$$1 + B_Y \equiv \frac{1 - Y_{\mathrm{A},\infty}}{1 - Y_{\mathrm{A},s}} \tag{52a}$$

$$B_Y = \frac{Y_{\mathrm{A},s} - Y_{\mathrm{A},\infty}}{1 - Y_{\mathrm{A},s}} \tag{52b}$$

Then the evaporation rate is

$$\dot{m} = 4\pi r_s \rho \mathcal{D}_\mathrm{AB} \ln(1 + B_Y) \tag{53}$$

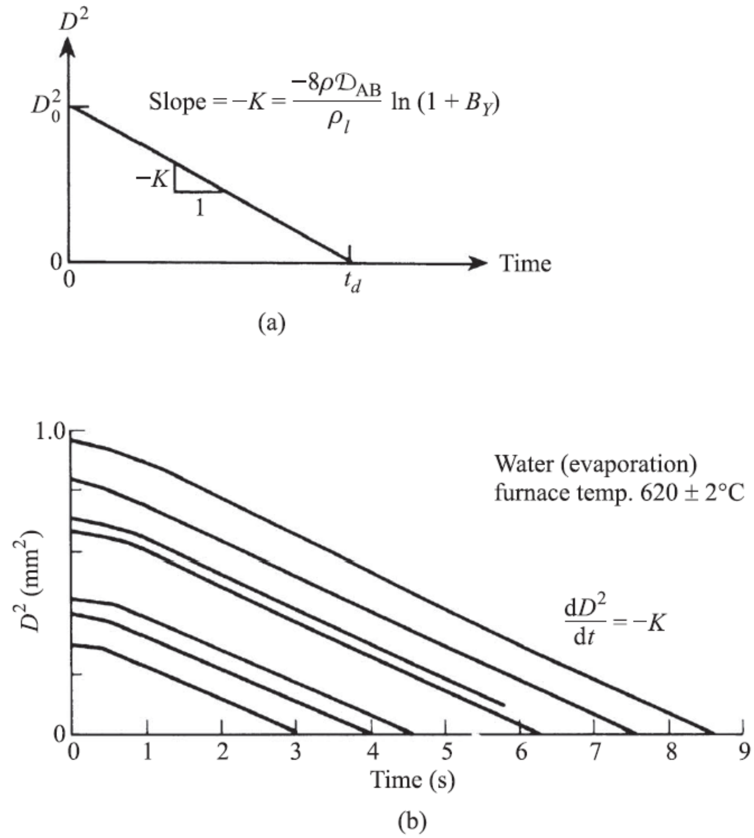

Figure 4: The D^2 law for droplet evaporation. (a) Simplified analysis. (b) Experimental data from Ref. [8] for water droplets with $T_\infty = 620^oC$.

Droplet Mass Conservation:

$$\frac{\mathrm{d}m_d}{\mathrm{d}t} = -\dot{m} \tag{54}$$

where m_d is given by

$$m_d = \rho_l V = \rho_l \pi D^3/6 \tag{55}$$

where $D = 2r_s$, and V is the volume of the droplet. Substituting (55) and (53) into (54) and differentiating

$$\frac{\mathrm{d}D}{\mathrm{d}t} = -\frac{4\rho\mathcal{D}_{\mathrm{AB}}}{\rho_l D}\ln\left(1+B_Y\right) \tag{56}$$

(56) is more commonly expressed in term of D^2 rather than D,

$$\frac{\mathrm{d}D^2}{\mathrm{d}t} = -\frac{8\rho\mathcal{D}_{\mathrm{AB}}}{\rho_l}\ln\left(1+B_Y\right) \tag{57}$$

(57) tells us that time derivative of the square of the droplet diameter is constant. D^2 varies with t with a slope equal to RHS of (57)7. This slope is defined as evaporation constant K:

$$K = \frac{8\rho\mathcal{D}_{\mathrm{AB}}}{\rho_l}\ln\left(1+B_Y\right) \tag{58}$$

Droplet evaporation time can be calculated from:

$$\int_{D_o^2}^{0} \mathrm{d}D^2 = -\int_{0}^{t_d} K\mathrm{d}t \tag{59}$$

which yields

$$t_d = D_o^2/K \tag{60}$$

We can change the limits to get a more general relationship to provide a general expression for the variation of D with time:

$$D^2(t) = D_o^2 - Kt \tag{61}$$

(61) is referred to as the D^2 law for droplet evaporation.